A Nine-Vertex Counterexample to Hypergraph Reconstruction: Disproof of Conjecture 00000001683

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Abstract

Two explicit simple 3-uniform hypergraphs on nine vertices have equal vertex-deleted decks, including multiplicities, but are not isomorphic. Neither is a Steiner triple system. Thus they refute both the assertion of reconstruction for every order at least eight and the assertion that every counterexample must consist of Steiner triple systems. The example is a classical construction of Kocay; the contribution here is an explicit, self-contained certificate and its kernel-checked Lean verification.

Reading of the source. Both source languages assert reconstruction for $n \geq 8$ and state that any counterexample must be a twin pair of Steiner triple systems. We use the ordinary vertex deck: the multiset of isomorphism classes of the induced hypergraphs obtained by deleting one vertex. A simple 3-uniform hypergraph on V is a set of three-element subsets of V. An isomorphism is a bijection of vertices carrying the edge set exactly onto the other edge set. A Steiner triple system requires every pair of distinct vertices to belong to exactly one edge. In particular, it is a 2-$(v, 3, 1)$ design; the source's phrase "3-design parameters" is not needed for the argument.

The explicit pair

Write abc for the unordered triple $\{a, b, c\}$. Let $V = \{0, 1, \ldots, 8\}$ and let the following twenty triples form B:

$$B = \{123, 128, 135, 136, 146, 147, 178, 234, 246, 247,\\ 257, 258, 345, 357, 358, 368, 456, 468, 567, 678\}.$$

Define

$$A_X = \{012, 016, 025, 034, 038, 047, 056, 078\},\\ A_Y = \{014, 018, 023, 027, 036, 045, 058, 067\},\\ E(X) = B \cup A_X, \qquad E(Y) = B \cup A_Y.$$

All listed triples have three distinct vertices, and within either edge set there are no repetitions. Thus both hypergraphs are simple, 3-uniform, and have 28 edges.

Theorem 1. *The hypergraphs X, Y have the same vertex deck but are not isomorphic. Neither X nor Y is a Steiner triple system. Consequently both clauses of the source described above are false.*

Proof. We prove the three assertions in turn.

Equal decks. The table gives permutations f of V. The entries in the last nine columns are

$f(0), \ldots, f(8)$. Each row is used for all deleted vertices in its first column.

deleted v	0	1	2	3	4	5	6	7	8
0	0	1	2	3	4	5	6	7	8
1, 5	0	1	8	7	6	5	4	3	2
2, 6	0	7	2	5	8	3	6	1	4
3, 7	0	5	8	3	6	1	4	7	2
4, 8	0	7	6	5	4	3	2	1	8

Each row is an involution and fixes the relevant deleted vertex v. Substitution into the displayed edge sets gives the following exact equalities:

v	$\#E(X - v) = \#E(Y - v)$	$f(E(X - v)) = E(Y - v)$
0	20	yes
1, 2, 7, 8	19	yes, using the row for each v
3, 4, 5, 6	18	yes, using the row for each v

These are finite equalities of sets of triples: apply the row permutation to every listed triple that avoids v, sort its entries, and compare with the other list after deleting triples containing v. They prove $X - v \cong Y - v$ for every v. The identity matching of the nine deleted vertices preserves all deck multiplicities.

Non-isomorphism. Let $d_H(a)$ count edges containing a, and let $c_H(a, b)$ count edges containing the distinct vertices a, b. An isomorphism preserves both quantities. The degree sequence in vertex order is, for both graphs,

$$\big(d_H(0), \ldots, d_H(8)\big) = (8, 9, 9, 10, 10, 10, 10, 9, 9).$$

Thus any isomorphism $e : X \longrightarrow Y$ must fix 0, preserve $L = \{1, 2, 7, 8\}$, and preserve $D = \{3, 4, 5, 6\}$. Since $c_X(1, 2) = 3$, the ordered pair $(e(1), e(2))$ must be one of

$$(1, 8), \quad (2, 7), \quad (7, 2), \quad (8, 1).$$

Indeed the only unordered pairs in L of Y with codegree three are $\{1, 8\}$ and $\{2, 7\}$. For the next step, the codegrees in Y between D and L are

c_Y	1	2	7	8
3	3	3	1	2
4	3	3	2	1
5	1	2	3	3
6	2	1	2	3

In X, we have

$$(c_X(3, 1), c_X(3, 2)) = (3, 2), \qquad (c_X(4, 1), c_X(4, 2)) = (2, 3).$$

The table therefore forces the additional image indicated in each row below. The last two columns give a contradiction to edge preservation.

$e(1)$	$e(2)$	forced image	T	$e(T)$	$T \in E(X)$	$e(T) \in E(Y)$
1	8	$e(3) = 3$	123	138	yes	no
2	7	$e(4) = 5$	124	257	no	yes
7	2	$e(4) = 4$	124	247	no	yes
8	1	$e(3) = 6$	123	168	yes	no

This exhausts all vertex bijections that could be isomorphisms.

Failure of the Steiner condition. In X, the pair $\{0, 1\}$ belongs to the two edges $012, 016$; in Y, it belongs to $014, 018$. In each case its codegree is two, whereas a Steiner triple system requires codegree one. Since $9 \geq 8$, the pair refutes reconstruction in the claimed range, and it also refutes the claim that counterexamples must be Steiner systems. □

The congruence clause. The order nine satisfies $9 \equiv 3 \pmod 6$. We do not dispute the usual necessary condition $v \equiv 1, 3 \pmod 6$ for a Steiner triple system. The counterexample invalidates the asserted restriction to Steiner systems, rather than that necessary condition itself. No meaning of "twin" can restore the assertion when neither member is Steiner.

Attribution

This is the $n = 3$ case, with eight ordinary vertices and the additional vertex 0, of Kocay's classical family [1]. Cooper and Okur [2], Definitions 9–13 and Theorem 2.1, describe the construction. We give all edges and verify this finite pair directly; we do not claim a new nonreconstruction construction. No theorem about the infinite family or characteristic polynomials is used in our proof.

Formal verification and scope

The Lean structure `Hypergraph3` is a Boolean relation on triples with proved symmetry under adjacent swaps and vanishing on repeated vertices. These properties express a simple 3-uniform hypergraph. The concrete objects `X` and `Y` use a bit indexed by $81a + 9b + c$ to represent the edge relation efficiently. The finite sets `edgesX` and `edgesY` contain exactly the unordered triples printed above. Their uniformity and cardinality 28 are proved. The bridge theorems

`edgeX_iff_mem`, `edgeY_iff_mem`

prove, for every $a, b, c \in \mathrm{Fin}(9)$, that the encoded edge relation holds if and only if $\{a, b, c\}$ belongs to the corresponding displayed edge set. Thus the bit encoding does not replace the mathematical hypergraph with an unrelated numerical assertion.

`IsIso` quantifies over every triple and preserves edges in both directions under an arbitrary vertex equivalence. `Hypomorphic` contains an outer bijection matching the deleted vertices and an inner vertex equivalence for each card; its restriction to the complements is an isomorphism. Conversely, any card isomorphism extends to such an equivalence by sending the omitted vertex to the omitted vertex. This is exactly equality of decks with multiplicity, not a weaker comparison of sets of card types. The finite equivalences and all card conditions are checked in `same_deck`.

`codegree` counts the actual third vertices completing a pair; `doubleDegree` counts ordered pairs completing a vertex and is twice its usual degree. Their preservation under arbitrary `IsIso` is proved by equivalences of the counting types. The theorem `not_isomorphic` excludes every vertex equivalence, using exhaustive finite cases pruned by these proved invariants and by individual edge contradictions.

The proposition `ReconstructionClaim` quantifies over every $n \geq 8$ and all simple 3-uniform hypergraphs on $\mathrm{Fin}(n)$. The final theorem

`conjecture1683_false : ¬ ReconstructionClaim`

is its negation. The theorems `X_not_steiner` and `Y_not_steiner` use the usual pair-codegree definition of a Steiner triple system. The theorem `steiner_clause_false` negates the universal claim that hypomorphic nonisomorphic pairs of order at least eight must both be Steiner systems. No statement about the arithmetic existence conditions for designs is formalised.

All finite checks use kernel evaluation with `decide`. The development contains no placeholder proofs, native decision procedures, or custom axioms. Only three standard axioms occur in the audit:

`propext`, `Classical.choice`, `Quot.sound`.

Source and reproduction

The exact bilingual source is included as `SOURCE.md`, from

`conjectures/00000001683.md`.

The source snapshot is pinned to upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

The project uses Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean`.

The supplied manifest pins the dependencies. Compile this document with Tectonic or a compatible LaTeX installation. No external numerical program is required to check the formal proof.

References

[1] W. L. Kocay, *A family of nonreconstructible hypergraphs*, Journal of Combinatorial Theory, Series B **42** (1987), 46–63.

[2] J. Cooper and U. Okur, *Polynomial Reconstruction Problem for Hypergraphs*, arXiv:2312.16152v2. Public construction and attribution.